Description of the 2017b Cross Section Matrix, v1

A. Kaboth (asher.kaboth@rhul.ac.uk)

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To evaluate splines, the "x" value of the spline is the value

$$x = \frac{w - w_{NEUT}}{w_{NEUT}} = \frac{w}{w_{NEUT}} - 1$$

for cases where $w_{NEUT} \neq 0$. In order to maintain consistency across evaluating the splines and calculating the priors for the cross section parameters, it is desirable to have the errors calculated as a fraction of the NEUT nominal, so that the (log of the) gaussian prior term can be calculated as

$$\left(\frac{x}{\sigma_{NEUT}}\right)^2$$

where $\sigma_{NEUT} = \frac{\sigma}{x_{NEUT}}$.

This section contains the 2017 NIWG errors calculated as such and accompanies the file This file contains:

xsec_cov The TMatrixT containing the covariance matrix

xsec_param_prior The TVectorD containing the prior means relative to the NEUT nominal

xsec_param_nom The TVectorD containing the NEUT nominal relative to itself—this is all 1s and 0s

xsec_param_prior_unnorm The TVectorD containing the prior means in their usual units (e.g., GeV/c^2 for M_As)

xsec_param_nom_unnorm The TVectorD containing the NEUT nominal in their usual units (e.g., GeV/c^2 for M_As)

xsec_param_lb The TVectorD containing the lower bounds for parameters relative to the NEUT nominal

xsec_param_ub The TVectorD containing the upper bounds for parameters relative to the NEUT nominal

xsec_param_names The names of the parameters

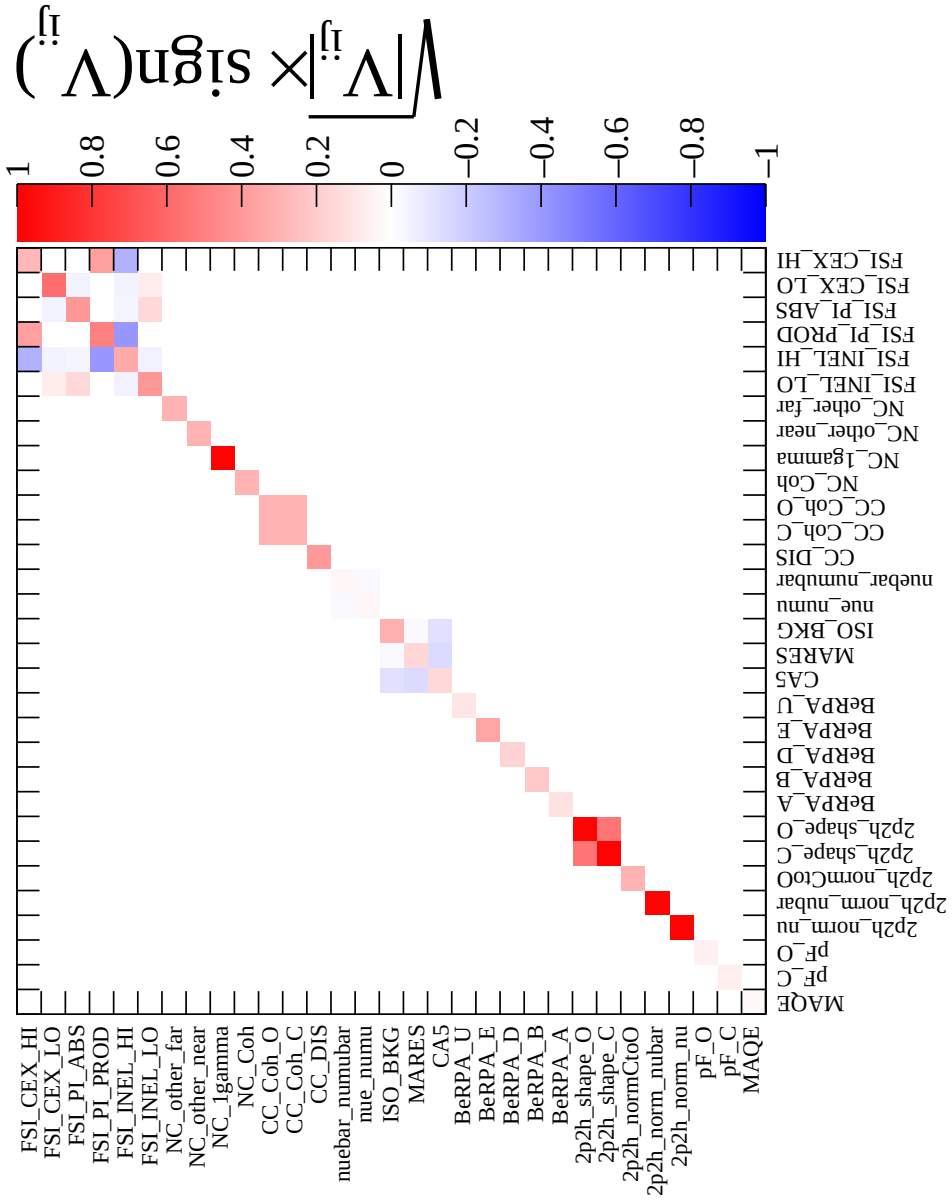

Figure 1: The 2017b covariance matrix, with the values shown as the sign of the element times the square root of the absolute value of the element.

Index	Parameter	Prior Mean	Error	NEUT nom	Prior/NEUT nom	Error/ NEUT nom	LB	UB
1	M_A^{QE}	1.20	0.03	1.20	1.00	0.03	0.00	9999.00
2	p_F C	217.00	13.00	217.00	1.00	0.06	0.92 (200)	1.27 (275)
3	p_F O	225.00	13.00	225.00	1.00	0.06	0.89 (200)	1.22 (275)
4	2p2h norm ν	1.00	1.00	1.00	1.00	1.00	0.00	9999.00
5	2p2h norm $\bar{\nu}$	1.00	1.00	1.00	1.00	1.00	0.00	9999.00
6	2p2h norm C to O	1.00	0.20	1.00	1.00	0.20	0.00	9999.00
7	2p2h shape C	0.00	3.00	0.00	0.00	3.00	-1.00	1.00
8	2p2h shape O	0.00	3.00	0.00	0.00	3.00	-1.00	1.00
9	BeRPA A	0.59	0.12	0.59	0.59	0.12	0.00	9999.00
10	BeRPA B	1.05	0.21	1.05	1.05	0.21	0.00	9999.00
11	BeRPA D	1.13	0.17	1.13	1.13	0.17	0.00	9999.00
12	BeRPA E	0.88	0.35	0.88	0.88	0.35	0.00	9999.00
13	BeRPA U	1.20	0.10	1.20	1.20	0.10	0.00	9999.00
14	CA5	0.96	0.15	1.01	0.95	0.15	0.00	9999.00
15	M_A^{RES}	1.07	0.15	0.95	1.13	0.16	0.00	9999.00
16	ISO BKG	0.96	0.40	1.30	0.74	0.31	0.00	9999.00
17	ν_e/ν_μ	1.00	0.03	1.00	1.00	0.03	0.00	9999.00
18	$\bar{\nu_e}/\bar{\nu_\mu}$	1.00	0.03	1.00	1.00	0.03	0.00	9999.00
19	CC DIS	0.00	0.40	0.00	0.00	0.40	-9999.00	9999.00
20	CC Coh C	1.00	0.30	1.00	1.00	0.30	0.00	9999.00
21	CC Coh O	1.00	0.30	1.00	1.00	0.30	0.00	9999.00
22	NC Coh	1.00	0.30	1.00	1.00	0.30	0.00	9999.00
23	NC 1γ	1.00	1.00	1.00	1.00	1.00	0.00	9999.00
24	NC other near	1.00	0.30	1.00	1.00	0.30	0.00	9999.00
25	NC other far	1.00	0.30	1.00	1.00	0.30	0.00	9999.00
26	FSI INEL LO	0.00	0.41	0.00	0.00	0.41	-9999.00	9999.00
27	FSI INEL HI	0.00	0.34	0.00	0.00	0.34	-9999.00	9999.00
28	FSI PI PROD	0.00	0.50	0.00	0.00	0.50	-9999.00	9999.00
29	FSI PI ABS	0.00	0.41	0.00	0.00	0.41	-9999.00	9999.00
30	FSI CEX LO	0.00	0.57	0.00	0.00	0.57	-9999.00	9999.00
31	FSI CEX HI	0.00	0.28	0.00	0.00	0.28	-9999.00	9999.00

Table 1: Parameters and relevant values of the 2017b covariance matrix. Parameters in red are typically used with flat priors in fits.